

PACKAGE INSERT – LEVOMED

1. NAME OF THE MEDICINAL PRODUCT

LEVOMED Tablet 25/250mg

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each tablet of LEVOMED Tablet 25/250mg contains 25mg carbidopa as carbidopa monohydrate and 250mg levodopa.

For the full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Tablet

LEVOMED Tablet 25/250mg: light blue, round, flat, mottled tablets, scored on one side, debossed “MC” on the other side with diameter 10.5mm

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

For treatment of Parkinson's disease and syndrome, particularly rigidity and bradykinesia. It is frequently helpful in the management of tremor, dysphagia, sialorrhoea, and postural instability associated with Parkinson's disease and syndrome.

4.2 Posology and method of administration

Method of administration

To be taken orally.

Posology

Dosage should be titrated to the individual patient needs and this may require adjusting both the individual dose and the frequency of administration. Each tablet of LEVOMED is designed to divide in half with minimal pressure.

Response has been observed in one day, and sometimes after one dose. Fully effective doses usually are reached within seven days as compared to weeks or months with Levodopa alone.

Studies show that the peripheral enzyme dopa decarboxylase is fully inhibited (saturated) by carbidopa at doses between 70 and 150mg daily. Patients receiving less than this amount of Carbidopa are more likely to experience nausea and vomiting.

Because both therapeutic and adverse responses occur more rapidly with LEVOMED than when levodopa is given, patients should be monitored closely during the dose adjustment period. Specifically, involuntary movements will occur more rapidly with LEVOMED than with Levodopa. The occurrence of involuntary movements may require dosage reduction. Blepharospasm may be useful early sign of excess dosage in some patients.

Patients Not Receiving Levodopa

One tablet 3x a day, if the patient remains uncontrolled, increase dose by one tablet every day or every other day until a maximum of 8 tablets.

Patients Receiving Levodopa

Levodopa should be discontinued at least 12 hours before LEVOMED is started (24 hours for slow-release preparations of Levodopa). A daily dosage of LEVOMED should be chosen that will provide approximately 20% of the previous Levodopa daily dosage.

Patients Receiving Levodopa and Another Decarboxylase Inhibitor

When transferring a patient to LEVOMED from Levodopa combined with another decarboxylase inhibitor, discontinue dosage at least 12 hours before LEVOMED is started. Begin with a dosage of LEVOMED that will provide the same amount of Levodopa as contained in the other Levodopa/decarboxylase inhibitor combination.

Patients Receiving Other Standard Antiparkinsonian Drugs

Current evidence indicates that other standard antiparkinsonian drugs may be continued although dosage may have to be adjusted.

4.3 Contraindications

Nonselective monoamine oxidase (MAO) inhibitors are contraindicated for use with LEVOMED. These inhibitors must be discontinued at least two weeks before starting LEVOMED. LEVOMED may be administered concomitantly with the manufacturer's recommended dose of an MAO inhibitor with selectivity for MAO type B (e.g. selegiline hydrochloride). (See section 4.5)

LEVOMED is contraindicated in patients with narrow-angle glaucoma and in patients with known hypersensitivity to any component of this medication.

Since levodopa may activate a malignant melanoma, it should not be used in patients with suspicious undiagnosed skin lesions or a history of melanoma.

Use during pregnancy or lactation and in women breast-feeding infants (see section 4.6).

Use in patients with severe psychoses.

4.4 Special warnings and precautions for use

LEVOMED is not recommended for the treatment of drug-induced extrapyramidal reactions.

LEVOMED should be administered cautiously to patients with severe cardiovascular or pulmonary disease, bronchial asthma, renal, hepatic or endocrine disease, or history of peptic ulcer disease (because of the possibility of upper gastro-intestinal haemorrhage).

Levodopa has been associated with somnolence and episodes of sudden sleep onset. Sudden onset of sleep during daily activities, in some cases without awareness or warning signs, has been reported very rarely. Patients must be informed of this and advised to exercise caution while driving or operating machines during treatment with levodopa. Patients who have experienced somnolence and/or an episode of sudden sleep onset must refrain from driving or operating machines. Furthermore a reduction of dosage or termination of therapy may be considered.

All patients should be monitored carefully for the development of mental changes, depression with concomitant suicidal tendencies, and other serious antisocial behaviour. Patients with past or current psychoses should be treated with caution.

Dyskinesias may occur in patients previously treated with levodopa alone because carbidopa permits more levodopa to reach the brain and, thus, more dopamine to be formed. The occurrence of dyskinesias may require dosage reduction.

Patients with a history of severe involuntary movements or psychotic episodes when treated with levodopa alone should be observed carefully when LEVOMED is substituted.

These reactions are thought to be due to increased brain dopamine following administration of levodopa, and use of LEVOMED may cause a recurrence. Dosage reduction may be required.

A syndrome resembling the neuroleptic malignant syndrome including muscular rigidity, elevated body temperature, mental changes and increased serum creatine phosphokinase has been reported with the abrupt withdrawal of antiparkinsonian agents. Therefore, patients should be observed carefully when the dosage of LEVOMED is reduced abruptly or discontinued especially if the patient is receiving neuroleptics.

Patients should be regularly monitored for the development of impulse control disorders. Patients and carers should be made aware that behavioural symptoms of impulse control disorders including pathological gambling, increased libido and hypersexuality compulsive spending or buying, binge eating and compulsive eating can occur in patients treated with dopamine agonists and/or other dopaminergic treatments containing levodopa including LEVOMED. Review of treatment is recommended if such symptoms develop.

If concomitant administration of psycho-active drugs such as phenothiazines or butyrophenones is necessary, such drugs should be administered with caution, and the patient carefully observed for loss of antiparkinsonian effect.

Patients with a history of convulsions should be treated with caution.

As with levodopa, periodic evaluation of hepatic, haematopoietic, cardiovascular and renal function is recommended during extended therapy.

Patients with chronic wide-angle glaucoma may be treated cautiously with LEVOMED, provided the intra-ocular pressure is well controlled and the patient monitored carefully for changes in intra-ocular pressure during therapy.

Care should be exercised when LEVOMED is administered to patients with a history of myocardial infarction who have residual atrial nodal, or ventricular arrhythmias. Cardiac function should be monitored with particular care in such patients during the period of initial dosage adjustment.

Melanoma: Epidemiological studies have shown that patients with Parkinson's disease have a higher risk (2- to approximately 6-fold higher) of developing melanoma than the general population. Whether the increased risk observed was due to Parkinson's disease or other factors, such as drugs used to treat Parkinson's disease, is unclear. For the reasons stated above, patients and providers are advised to monitor for melanomas frequently and on a regular basis when using LEVOMED for any indication. Ideally, periodic skin examinations should be performed by appropriately qualified individuals (e.g. dermatologists).

If general anaesthesia is required, therapy with LEVOMED may be continued for as long as the patient is permitted to take fluids and medication by mouth. If therapy is interrupted temporarily, the usual daily dosage may be administered as soon as the patient is able to take oral medication.

Laboratory Tests

Transient abnormalities in laboratory test results may occur, but have not been associated with clinical evidence of disease. These include elevated levels of blood urea nitrogen, AST (SGOT), ALT (SGPT), LDH, bilirubin, creatinine, uric acid and alkaline phosphatase.

Commonly, levels of blood urea nitrogen, creatinine, and uric acid are lower during administration of LEVOMED than with levodopa.

Decreased haemoglobin, haematocrit, elevated serum glucose and white blood cells, bacteria and blood in the urine have been reported.

Positive Coombs' tests have been reported, both with LEVOMED and levodopa alone, but haemolytic anaemia is extremely rare.

LEVOMED may cause a false positive result for ketonuria when a dipstick is used to test for urinary ketone bodies; and this reaction is not altered by boiling the urine specimen. The use of glucose oxidase methods may give false negative results for glycosuria.

4.5 Interaction with other medicinal products and other forms of interaction

Caution should be exercised when the following drugs are administered concomitantly with LEVOMED.

Antihypertensive agents

Symptomatic postural hypotension has occurred when LEVOMED is added to the treatment of patients receiving some antihypertensive drugs. Therefore, when therapy with LEVOMED is started, dosage adjustment of the antihypertensive drug may be required.

Antidepressants

Rarely, reactions including hypertension and dyskinesia have been reported with the concomitant use of tricyclic antidepressants. (For patients receiving monoamine oxidase inhibitors, see first paragraph of section 4.3).

Iron

Studies demonstrate a decrease in the bioavailability of carbidopa and/or levodopa when it is ingested with ferrous sulphate or ferrous gluconate.

Other drugs

Dopamine D2 receptor antagonists (e.g. phenothiazines, butyrophenones, and risperidone, thioxanthenes) and isoniazid may reduce the therapeutic effects of levodopa. The beneficial effects of levodopa in Parkinson's disease have been reported to be reversed by phenytoin and papaverine. Patients taking these drugs with LEVOMED should be carefully observed for loss of therapeutic response.

Use of LEVOMED with dopamine-depleting agents (e.g. reserpine and tetrabenazine) or other drugs known to deplete monoamine stores is not recommended.

Concomitant therapy with selegiline and carbidopa-levodopa may be associated with severe orthostatic hypotension not attributable to carbidopa-levodopa alone (See section 4.3)

Since levodopa competes with certain amino acids, the absorption of LEVOMED may be impaired in some patients on a high protein diet.

The effect of simultaneous administration of antacids with LEVOMED on the bioavailability of levodopa has not been studied.

4.6 Fertility, pregnancy and lactation

Pregnancy

Although the effects of LEVOMED on human pregnancy are unknown, both levodopa and combinations of carbidopa and levodopa have caused visceral and skeletal malformations in rabbits. Therefore, the use of LEVOMED in women of childbearing potential requires that the anticipated benefits of the drug be weighed against possible hazards should pregnancy occur.

Use of LEVOMED is contraindicated in pregnancy.

Breast-feeding

It is not known whether carbidopa is excreted in human milk. In a study of one nursing mother with Parkinson's disease, excretion of levodopa in breast milk was reported. Because many drugs are excreted in human milk and because of the potential for serious adverse reactions in infants, a decision should be made whether to discontinue breast-feeding or discontinue the use of LEVOMED, taking into account the importance of the drug to the mother.

4.7 Effects on ability to drive and use machines

No data are known about the effect on the ability to drive. If side effects such as dizziness or somnolence occur, they may affect the ability to drive and to operate machinery.

Patients being treated with levodopa and presenting with somnolence and/or sudden sleep episodes must be informed to refrain from driving or engaging in activities where impaired alertness may put themselves or others at risk of serious injury or death (e.g. operating machines), until such recurrent episodes and somnolence have resolved (see also section 4.4).

4.8 Undesirable effects

Side effects that occur frequently with LEVOMED are those due to the central neuropharmacological activity of dopamine. These reactions can usually be diminished by dosage reduction. The most common are dyskinesias including choreiform, dystonic and other involuntary movements and nausea. Muscle twitching and blepharospasm may be taken as early signs to consider dosage reduction.

Other side effects reported in clinical trials or in post-marketing experience include:

Metabolism and nutrition disorders:

Common: Anorexia

Psychiatric disorders:

Common: Dream abnormalities, hallucinations, depression with or without development of suicidal tendencies, confusion.

Uncommon: Agitation.

Rare: Psychotic episodes including delusions and paranoid ideation, increased libido.

Nervous system disorders:

Common: Bradykinetic episodes (the "on-off" phenomenon), dizziness, somnolence including very rarely excessive daytime somnolence and sudden sleep onset, paresthesia.

Uncommon: Syncope

Rare: Dementia, neuroleptic malignant syndrome (see 4.4 'Special warnings and precautions for use'). Rarely convulsions have occurred; however, a causal relationship with LEVOMED has not been established.

Cardiac disorders:

Common: Palpitation.

Rare: Cardiac irregularities.

Vascular disorders:

Common: Orthostatic effects including hypotensive episodes.

Rare: Hypertension, phlebitis.

General disorders and administration site conditions:

Common: Chest pain.

Gastrointestinal disorders:

Common: Diarrhoea, vomiting.

Rare: Gastrointestinal bleeding, development of duodenal ulcer, dark saliva.

Blood and lymphatic system disorders:

Rare: Leukopenia, haemolytic and non-haemolytic anaemia, thrombocytopenia, agranulocytosis.

Respiratory, thoracic and mediastinal disorders:

Common: Dyspnoea.

Skin and subcutaneous disorders:

Uncommon: Urticaria.

Rare: Angioedema, pruritis, Henoch-Schönlein purpura, alopecia, rash, dark sweat.

Renal and urinary disorders:

Rare: Dark urine.

Other side effects that have been reported with levodopa or levodopa/carbidopa combinations and may be potential side effects with LEVOMED include:

Neoplasms, benign, malignant and unspecified (including cysts and polyps):

Malignant melanoma (see section 4.3).

Gastro-intestinal disorders:

Dry mouth, sialorrhoea, dysphagia, hiccups, dyspepsia, abdominal pain and distress, constipation, flatulence, burning sensation of the tongue.

Metabolism and nutrition disorders:

Weight gain or loss.

Psychiatric disorders:

Anxiety, disorientation, euphoria, insomnia, bruxism.

Nervous system disorders:

Bitter taste, decreased mental acuity, ataxia, numbness, increased hand tremor, activation of latent Horner's syndrome, extrapyramidal and movement disorders, faintness, headache, sense of stimulation. Impulse control disorders: Pathological (compulsive) gambling, increased libido, hypersexuality, compulsive spending or buying, binge eating and compulsive eating can occur in patients treated with dopamine agonists and/or other dopaminergic treatments containing levodopa including LEVOMED (see section 4.4).

Skin and subcutaneous disorders:

Increased sweating.

Musculoskeletal, connective tissue and bone disorders:

Muscle cramps, trismus.

Eye disorders:

Diplopia, blurred vision, dilated pupils, oculogyric crises.

Vascular disorders:

Hot flashes, flushing.

Respiratory, thoracic and mediastinal disorders:

Hoarseness, bizarre breathing patterns.

Renal and urinary disorders:

Urinary retention, urinary incontinence.

Reproductive system and breast disorders:

Priapism.

General disorders:

Weakness, asthenia, fatigue, malaise, oedema, gait abnormalities.

Injury, poisoning and procedural complications:

Falling.

4.9 OverdoseTreatment

Management of acute overdosage with LEVOMED is basically the same as management of acute overdosage with levodopa; however pyridoxine is not effective in reversing the actions of LEVOMED. ECG monitoring should be instituted, and the patient carefully observed for the possible development of arrhythmias; if required, appropriate anti-arrhythmic therapy should be given. The possibility that the patient may have taken other drugs as well as LEVOMED should be taken into consideration.

To date, no experience has been reported with dialysis, and hence its value in the treatment of overdosage is not known.

5. PHARMACOLOGICAL PROPERTIES**5.1 Pharmacodynamic properties**

Pharmacotherapeutic group: Dopa and dopa derivatives, levodopa and decarboxylase inhibitor, ATC Code: N04BA02.

Levodopa is a precursor of dopamine, and is given as replacement therapy in Parkinson's disease.

Carbidopa is a peripheral dopa decarboxylase inhibitor. It prevents metabolism of levodopa to dopamine in the peripheral circulation, ensuring that a higher proportion of the dose reaches the brain, where dopamine acts. A lower dose of levodopa can be used, reducing the incidence and severity of side effects.

5.2 Pharmacokinetic properties

Following oral dosing levodopa, in the absence of decarboxylase inhibitor, is rapidly but variably absorbed from the gastrointestinal tract. It has a plasma half-life of about 1 hour and is mainly converted by decarboxylation to dopamine, a proportion of which is converted to noradrenaline. Up to 30 % is converted to 3-O-methyldopa which has a half-life of 9 to 22 hours. About 80 % of levodopa is excreted in the urine within 24 hours mainly as homovanillic acid and dihydroxyphenylactic acid. Less than 1 % is excreted unchanged.

Once in the circulation it competes with other neutral amino acids for transport across the blood brain barrier. Once it has entered the striatal neurones it is decarboxylated to dopamine, stored and released from presynaptic neurones. Because levodopa is so rapidly decarboxylated in the gastrointestinal tract and the liver, very little unchanged drug is available for transport into the brain. The peripheral decarboxylation reduces the therapeutic effectiveness of levodopa but is responsible for many of its side effects. For this reason levodopa is usually administered together with a peripheral decarboxylase inhibitor such as carbidopa, so that lower doses may be given to achieve the same therapeutic effect.

Carbidopa in the absence of levodopa, is rapidly but incompletely absorbed from the gastrointestinal tract following oral dosing. Following an oral dose approximately 50% is recorded in the urine, with about 30 % of this as unchanged drug. It does not cross the blood brain barrier but crosses the placenta and is excreted in breast milk. Turnover of the drug is rapid and virtually all unchanged drug appears in the urine within 7 hours.

Carbidopa inhibits the peripheral decarboxylation of levodopa to dopamine but as it does not cross the blood brain barrier, effective brain levels of dopamine get produced with lower levels of levodopa therapy reducing the peripheral side effects, noticeably nausea and vomiting and cardiac arrhythmias.

5.3 Preclinical safety data

Not applicable

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

LEVOMED Tablet 25/250mg: Pregelatinised starch, Microcrystalline cellulose PH101, Microcrystalline cellulose PH102, Crospovidone type B, Hydroxypropylcellulose, Indigo carmine (E132), Magnesium stearate.

6.2 Incompatibilities

Not applicable.

6.3 Shelf life

36 months

6.4 Special precautions for storage

Store below 30°C in the original package, in order to protect from light.

6.5 Nature and contents of container

Alu-Alu blister packs 20 and 100 tablets

6.6 Special precautions for disposal

Not applicable.

7. MARKETING AUTHORISATION HOLDER

Komedic Sdn. Bhd., No. 4 Jalan PJS 11/14, Bandar Sunway, 46150 Petaling Jaya, Selangor, Malaysia

8. MANUFACTURER

Medochemie Ltd.,(Central Factory) Konstantinoupoleos 1-10, Limassol, 3011, Cyprus

9. MARKETING AUTHORISATION NUMBER(S)

MAL19900635AZ

10. DATE OF REVISION OF THE TEXT

05/26